Predicting Human Translation Difficulty Using Automatic Word Alignment

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Motivation

1. **Hard-to-translate words** often go beyond simple one-to-one correspondences, and can be **challenging to translators** because of cultural context or differences in semantic subdivisions.
2. Understanding translation difficulty is an important scientific challenge in its own right, but could have other **practical applications**, e.g.,
   - Developing targeted evaluations for NMT systems
   - Generation of translation samples of varying difficulties to facilitate human evaluation of MT

Heart
心臓 (shinzō, the organ)
心 (kokoro, mind/spirit)
ハート (hāto, borrowed)

Fuel
燃料 (nenryō),
ガソリン(gàsóríń, borrowed)

Methodology

Fuel is *easier* to translate to Japanese than **heart**.

Alignment-based predictors

1. Word alignment probability

$$p_{al}(v|w) = \frac{S_{w \leftrightarrow v}}{\sum_{u \in V} S_{w \leftrightarrow u}}$$

2. Translation surprisal

$$I_{al}(v|w) = -\log p_{al}(v|w)$$

3. Translation uncertainty

$$^{l}H_{al}(w) = -\sum_{u \in V} p_{al}(u|w) \log p_{al}(u|w)$$

Human translation difficulty

Context-free translations. Translation norms from **psycholinguistic** literatures: *number of translations, semantic similarity ratings*

Context-dependent translations. Human **translation processing** data: *CRITT TPR-DB*

	Session	SL	TL	Task	SToken	TGroup	Dur	Munit	HTra
0	P02_T3	en	ja	T	Spielberg	スピルバーグ_、	165986	7	3.4098
1	P02_T3	en	ja	T	shows	を_動告	827	3	3.7802
2	P02_T3	en	ja	T	Beijing	中国_に	7566	5	2.9134

Baseline

Thompson et al.'s (2020) embedding-based approach (S_{emb} / M_{emb})
word-pair semantic alignments using Common Crawl and Wikipedia fastText embeddings

Results

heart → 3 translations (ja)
fuel → 2 translations (ja)

		es	ja	ms	nl	zh
→ en	M_{emb}	.300	.341	-	.247	-
	H^c_{al}	.442	.563	.255	.264	-
	H^w_{al}	**.451**	**.570**	**.266**	**.270**	-
en →	M_{emb}	.351	.461	-	**.358**	.284
	H^c_{al}	**.487**	.525	.430	.250	.348
	H^w_{al}	**.487**	**.538**	**.440**	.248	**.351**

Context-free translations:
Pearson correlation ($p < .001$) between alignment distribution entropy and *number of translations* in translation norms.

rating(***heart***, ハート) = 3.7
rating(***fuel***, 燃料) = 4.5

		ja	nl	zh
→ en	S_{emb}	**-.422**	-.302	-.332
	I^c_{al}	-.200	**-.587**	**-.474**
	I^w_{al}	-.194	**-.587**	-.471
en →	S_{emb}	-.422	-.284	-.332
	I^c_{al}	**-.474**	**-.476**	**-.486**
	I^w_{al}	.-.471	-.474	-.484

Context-free translations:
Pearson correlation ($p < .01$) with *word-pair similarity ratings*.

Dur(***heart***) = 7.69 ms
Dur(***fuel***) = 7.50 ms

		de	es	ja
HTra↑	M_{emb}	.322	.298	.273
	H^c_{al}	.427	**.512**	**.406**
	H^w_{al}	**.428**	.511	.405
Dur (ms)↑	S_{emb}	-.363	-.466	/
	I^c_{al}	**-.109**	**-.195**	-.161
	I^w_{al}	-.120	-.205	**-.156**

Context-dependent translations:
Correlations ($p < .05$) between alignment distribution entropy and translation processing data.

Discussion

Lexical and Morphological Variation

- Word-alignment approach more accurately **captures the distribution of lexical choices and morphological variants**
- Poor representations of rare and morphologically complex words in static and contextual embeddings
- Word embeddings typically optimized to minimize the contribution of frequency information

easier to translate →

Effects of Form Similarity

- **Cognate status** of a source word increases translation duration
- Context helps facilitate non-cognate alternatives to compete in lexical selection

en	es	Dur	S_{emb}	I^c_{al}	I^w_{al}
disliked	*disgustaba*	5.90	-.555	2.79	2.71
	detestaba	8.53	-.555	4.45	4.37
region	*región*	8.01	-.393	0.21	0.20
	zona	7.36	-.391	2.74	2.73
varied	*variada*	6.70	-.587	1.66	1.64
	variado	7.15	-.628	2.05	2.02
	diversa	7.11	-.600	5.76	5.73
	diverso	6.95	-.586	5.76	5.73

Conclusion

Compared to the embedding-based approach, our measures **improved estimates of translation difficulty**, but suggest that a comprehensive account of human translation difficulty must also consider additional factors such as form similarity.

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